

Key Decisions

Topic Decision	Decision
Object-oriented programming language (if not using OOP, then justify the use of the language)	Typescript
GUI Library	React
IDE	Visual Studio Code
xUnit Framework	Vitest / Jest
Programming Style Guide	Google Typescript Style Guide
Project Hosting Site	Github

Unit Testing

Original Script Output:

```
PS C:\Users\tomie\General\Github\React-Tutorial-Project> npx tsx .\src\script.ts\  
add(1, 2) = 3
```

Original Script source code:

```
src > TS script.ts > ...  
1 export function add(a: number, b: number) {  
2   return a + b;  
3 }  
4  
5 console.log("add(1, 2) =", add(1, 2));|
```

Unit Test Source Code:

```

JS script.test.js > ...
import { describe, it, expect } from "vitest";
import { add } from "../src/script.ts";

describe("add function", () => {
  it("should return 3 when adding 1 and 2", () => {
    expect(add(1, 2)).toBe(3);
  });

  it("should handle negative numbers correctly", () => {
    expect(add(-1, -5)).toBe(-6);
  });
});

```

Unit Test Output:

```

RERUN src/script.ts x4

stdout | test/script.test.js
add(1, 2) = 3

✓ test/script.test.js (2 tests) 7ms
  ✓ add function (2)
    ✓ should return 3 when adding 1 and 2 4ms
    ✓ should handle negative numbers correctly 0ms

Test Files  1 passed (1)
Tests       2 passed (2)
Start at    18:24:39
Duration    56ms

PASS Waiting for file changes...

```

GUI Programming

Requirements: text, lines, a check box, and radio buttons.

Habit Tracker

Sep 14 - Sep 20

PrevNext

New habit...Add Habit

Habit 1🔥 1Delete

Mon14

Tue15

Wed16

Thu17

Fri18

Sat19

Sun20

Habit 2🔥 1Delete

Mon14

Tue15

Wed16

Thu17

Fri18

Sat19

Sun20

Habit 3🔥 1Delete

Mon14

Tue15

Wed16

Thu17

Fri18

Sat19

Sun20

Note: None of the below is functional or connected to anything above. This is purely to fulfill the assignment requirements.

Select View

☒ All☐ Completed☐ Pending

Source code repository link: [CallmeCheese1/React-Tutorial-Project: My first project to learn react, based on Web Dev Simplified's "Learn React With This One Project" video.](#)

To be specific, [App.tsx](#).